

Literature Review

Introduction

The exponential growth of structured data in relational databases has intensified the need for intuitive data access methods. Traditional SQL expertise remains a significant barrier for non-technical users, creating operational inefficiencies. Large Language Models (LLMs) offer transformative potential in bridging this gap by translating natural language queries into SQL. This review synthesizes research on LLM-based Text-to-SQL systems, focusing on architectural frameworks, model performance, evaluation benchmarks, and real-world implementation challenges.

Methodology

The search was conducted using the following keywords: **“Text-to-SQL transformer models”**, **“LLM for SQL generation”**, **“Spider dataset”**. Databases used included **“PubMed”** and **“Google Scholar”**. The research was not filtered by date or type.

Inclusion and Exclusion Criteria

In conducting the literature review, inclusion and exclusion criteria were applied to ensure that only studies directly relevant to Text-to-SQL were considered. The criteria were defined to capture works that provide methodological, evaluative, or application-driven contributions to the field, while excluding papers that focus broadly on language models or theoretical aspects without a clear connection to Text-to-SQL.

Inclusion Criteria:

Studies were included if they:

- Focused on **methods, systems, or frameworks for translating natural language to SQL**.
- Evaluated **LLM-based Text-to-SQL approaches** on benchmark or domain-specific datasets.
- Introduced **new datasets, benchmarks, or evaluation methods** for Text-to-SQL.
- Presented **survey papers** dedicated to Text-to-SQL.
- Applied Text-to-SQL in **specific domains or real-world scenarios** (e.g., healthcare, IoT, software engineering).

Exclusion Criteria:

Studies were excluded if they:

- Addressed **general LLM topics** (e.g., scaling laws, fine-tuning) without connection to Text-to-SQL.
- Focused on tasks **other than natural language to SQL generation**, such as SQL debugging, SQL equivalence checking, or general query optimization.
- Concentrated on **theoretical AI concepts** with no relevance to SQL generation.
- Focused on **multimodal, vision, or unrelated NLP tasks**.

No	Article	Authors	Published	Link	Keywords	Conclusions
1	Next-Generation Database Interfaces: A Survey of LLM-based Text-to-SQL	Zijin Hong, Zheng Yuan, Qinggang Zhang, Hao Chen, Junnan Dong, Feiran Huang, Xiao Huang	2025	Link	Text-to-sql, LLM	This survey reviews how large language models have transformed text-to-SQL tasks, highlighting key techniques, datasets, and challenges. It emphasizes recent LLM-based methods that improve query accuracy and discusses future directions for making these systems more robust and explainable.
2	From Natural Language to SQL: Review of LLM-based Text-to-SQL Systems	Ali Mohammadjafari, Anthony S. Maida, Raju Gottumukkala	2025	Link	Text-to-sql, LLM	The paper surveys advances in translating natural language queries into SQL using large language models, tracking progress from early rule-based systems to modern LLM-powered solutions, and underlining their impact on database accessibility.
3	SQLfuse: Enhancing Text-to-SQL Performance through Comprehensive LLM Synergy	Tingkai Zhang, Chaoyu Chen, Cong Liao, Jun Wang, Xudong Zhao, Hang Yu, Jianchao Wang, Jianguo Li, Wenhui Shi	2024	Link	Text-to-sql, LLM	Text-to-SQL simplifies database querying by enabling users to ask questions in natural language. This approach expands access to data analysis, making it easier for non-experts to interact with complex databases.
4	Decomposition for Enhancing Attention: Improving LLM-based Text-to-SQL through Workflow Paradigm	Yuanzhen Xie, Xinzhou Jin, Tao Xie, MingXiong Lin, Liang Chen, Chenyun Yu, Lei Cheng, ChengXiang Zhuo, Bo Hu, Zang Li	2024	Link	Text-to-sql, LLM	To overcome limitations in single-step LLM prompting for text-to-SQL, the authors propose a workflow combining decomposition, self-correction, and active learning. This method enhances LLM performance and sets new benchmarks on standard datasets.
5	Knowledge-to-SQL: Enhancing SQL Generation with Data Expert LLM	Zijin Hong, Zheng Yuan, Hao Chen, Qinggang Zhang, Feiran Huang, Xiao Huang	2024	Link	Text-to-sql, LLM	The paper introduces a framework that supplements missing domain knowledge in database schemas with a specialized language model, improving the accuracy and robustness of text-to-SQL generation beyond what standard LLMs achieve.
6	LLM-Based Text-to-SQL for Real-World Databases	Eduardo R. Nascimento, Grettel García, Yenier T. Izqueirdo, Lucas Feijó, Gustavo M. C. Coelho, Aiko R. de Oliveira, Melissa Lemos, Robinson L. S. Garcia, Luiz A. P. Paes Leme & Marco	2024	Link	Text-to-sql, LLM	This study reveals that existing LLM-based text-to-SQL tools struggle with complex real-world databases due to schema representation issues. It proposes LLM-friendly views and data descriptions aligned with user language to significantly improve query generation performance.

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7	Text-to-SQL Empowered by Large Language Models: A Benchmark Evaluation	Dawei Gao, Haibin Wang, Yaliang Li, Xiuyu Sun, Yichen Qian, Bolin Ding, Jingren Zhou	2023	Link	Text-to-sql, LLM	The authors systematically compare various prompt engineering techniques for LLM-based text-to-SQL systems—examining question representation, example selection, and organization strategies. Based on these insights, they introduce DAIL-SQL , which achieves a new state-of-the-art execution accuracy of 86.6 % on the Spider benchmark
8	Knapsack Optimization-based Schema Linking for LLM-based Text-to-SQL Generation	Zheng Yuan, Hao Chen, Zijin Hong, Qinggang Zhang, Feiran Huang, Qing Li, Xiao Huang	2025	Link	Text-to-sql, LLM	This paper addresses the critical schema linking step in text-to-SQL systems, which often suffers from missing relevant tables or including redundant ones. The authors introduce KaSLA , a plug-in method using knapsack optimization and a hierarchical linking strategy to accurately select tables and columns. Their enhanced schema linking metric and approach significantly boost SQL generation performance on benchmarks like Spider and BIRD, outperforming prior state-of-the-art models
9	TURSpider: A Turkish Text-to-SQL Dataset and LLM-Based Study	Ali Bugra Kanburoglu, Faik Boray Tek	2024	Link	Test-to-sql, LLM, Spider	This paper presents TURSpider, the first large-scale Turkish Text-to-SQL dataset, adapted from Spider. It benchmarks LLMs for Turkish SQL generation, introducing a feedback mechanism and a multi-agent approach (MoAF-SQL) to improve performance and robustness.
10	Large Language Model Enhanced Text-to-SQL Generation: A Survey	Xiaohu Zhu, Qian Li, Lizhen Cui, Yongkang Liu	2024	Link	Sql, LLM, Survey	The authors provide a comprehensive survey of text-to-SQL generation approaches enhanced by LLMs, categorizing methods into prompt engineering, fine-tuning, task-specific training, and agent-based systems. They also offer a detailed review of datasets and evaluation metrics. This work aims to clarify current research patterns, identify challenges, and guide future developments in LLM-powered text-to-SQL systems
11	A Survey on Employing Large Language Models for Text-to-SQL Tasks	Liang Shi, Zhengju Tang, Nan Zhang, Xiaotong Zhang, Zhi Yang	2025	Link	Sql, LLM, Survey	This survey reviews over 170 studies on using LLMs for text-to-SQL, comparing prompting and fine-tuning methods. It outlines key challenges like schema linking and domain adaptation, offering guidance for future research
12	A Survey of Large Language Model-Based Generative AI for Text-to-SQL: Benchmarks, Applications, Use Cases, and Challenges	Aditi Singh, Akash Shetty, Abul Ehtesham, Saket Kumar, Tala Talaei Khoei	2025	Link	Sql, LLM, Survey	This survey reviews recent LLM-based approaches to text-to-SQL, categorizing works by prompt engineering, fine-tuning, task-specific training, and agent-based methods. It synthesizes insights from over 170 studies, highlights prevailing challenges such as schema linking and domain adaptation, and outlines future research directions in efficiency, accuracy, and robustness
13	A Survey of Text-to-SQL in the Era of LLMs: Where are We, and Where are We Going?	Xinyu Liu, Shuyu Shen; Boyan Li, Peixian Ma, Runzhi Jiang, Yuxin Zhang	2025	Link	Sql, LLM, Survey	This survey reviews LLM-based text-to-SQL systems, presenting a modular taxonomy, error classification, and practical guidance. It highlights key challenges like generalization, efficiency, and robustness for future work.

14	Exploring the Landscape of Text-to-SQL with Large Language Models: Progresses, Challenges and Opportunities	Yiming Huang, Jiyu Guo, Wenxin Mao, Cuiyun Gao, Peiyi Han, Chuanyi Liu, Qing Ling	2025	Link	Sql, LLM	This systematic survey examines LLM-based text-to-SQL research, focusing on trends, techniques, datasets, evaluation metrics, and challenges. It offers a clear roadmap for future work by highlighting gaps and opportunities in schema linking, error analysis, and domain adaptation
15	Enhancing LLM Fine-tuning for Text-to-SQLs by SQL Quality Measurement	Shouvon Sarker, Xishuang Dong, Xiangfang Li, Lijun Qian	2024	Link	Sql, LLM	The authors propose a novel feedback mechanism that evaluates generated SQL based on quality and execution results, then uses this to fine-tune LLMs. Tested on the BIRD benchmark, it achieves execution accuracy and efficiency comparable to state-of-the-art models like GPT-4 and T5
16	Optimizing text-to-SQL conversion techniques through the integration of intelligent agents and large language models	Samuel Ojuri , The Anh Han , Raymond Chiong , Alessandro Di Stefano	2023	Link	Sql, LLM	This study combines intelligent agents with LLMs to help non-technical users generate accurate SQL queries. Fine-tuned examples significantly improve performance, enabling faster and easier data access.
17	Conversion of natural language text to SQL queries using generative AI	Arnav Jha, Naman Anand, H. Karthikeyan	2025	Link	Sql, LLM	This work introduces Semantic Synthesis , a method that converts user-provided natural language into precise SQL queries using a deep integration of LLMs and transformers. It targets non-technical users by providing context-aware SQL generation across domains like healthcare, e-commerce, education, and HR, paired with algorithmic transparency for the translation process. It aims to democratize database access via an intuitive interface that balances accuracy and interpretability
18	Text-to-SQL Domain Adaptation via Human-LLM Collaborative Data Annotation	Yuan Tian, Daniel Lee, Fei Wu, Tung Mai, Kun Qian, Siddhartha Sahai, Tianyi Zhang, Yunyao Li	2025	Link	Text-to-sql, LLM,survey	The paper introduces SQLsynth , a human-in-the-loop annotation system that combines experts and LLMs to efficiently generate domain-specific text-to-SQL training data. In user studies, SQLsynth reduces cognitive load, speeds up annotation, and produces more accurate and diverse datasets compared to manual annotation or using ChatGPT alone
19	LLM driven Natural Language to SQL for Utility Dashboards	Jyotirmoy Banerji; Abhiraj Amin; Kapil Rathor	2025	Link	Sql, LLM	This paper introduces an NL2SQL system optimized for IoT plant data, using fine-tuned LLMs and custom context structures. It improves SQL generation accuracy for complex, context-rich IoT queries and is evaluated with the Spider dataset.
20	Analyzing the Effectiveness of Large Language Models on Text-to-SQL Synthesis	Richard Roberson, Gowtham Kaki, Ashutosh Trivedi	2024	Link	Sql, LLM, Survey	This study compares fine-tuning an open-source model (WizardCoder-15B) with a pipeline using GPT-3.5-Turbo plus GPT-4 for error correction. WizardCoder achieves ~61% execution accuracy, while the GPT-based pipeline reaches ~82%. The authors analyze common mistakes in generated SQL—such as wrong columns, join errors, and misordered aggregations—offering insights for future improvements.
21	SWE-SQL: Illuminating LLM Pathways to Solve	Jinyang Li, Xiaolong Li, Ge Qu, Per Jacobsson, Bowen	2025	Link	Sql, LLM	This work introduces BIRD-CRITIC , a new benchmark comprising 530 PostgreSQL-only and 570 multi-dialect SQL debugging tasks

	User SQL Issues in Real-World Applications	Qin, Binyuan Hui, Shuzheng Si, Nan Huo, Xiaohan Xu, Yue Zhang, Ziwei Tang, Yuanshuai Li, Florensia Widjaja, Xintong Zhu, Feige Zhou, Yongfeng Huang, Yannis Papakonstantinou, Fatma Ozcan, Chenhao Ma, Reynold Cheng				sourced from real-world user issues. Standard LLMs achieve only ~39% success. To tackle this, the authors propose Six-Gym —an environment that reverse-engineers SQL problems (SQL-Rewind) and leverages f-Plan Boosting to train models more effectively. Their agent Bird-Fixer , built on Qwen-2.5-Coder-14B, outperforms leading proprietary models on realistic SQL-debugging tasks
22	LLM SQL Solver: Can LLMs Determine SQL Equivalence?	Fuheng Zhao, Jiayue Chen, Lawrence Lim, Ishtiyaque Ahmad, Divyakant Agrawal, Amr El Abbadi	2023	Link	Sql, LLM	This paper explores using LLMs to judge if two SQL queries are semantically equivalent. It proposes prompt-based methods that show strong alignment with human judgment, offering a promising alternative to traditional evaluation metrics.
23	Improving Text-to-SQL with a Hybrid Decoding Method	Geunyeong Jeong , Mirae Han , Seulgi Kim , Yejin Lee , Joosang Lee , Seongsik Park , Harksoo Kim	2023	Link	Text-to-sql	This study proposes a hybrid decoder that combines sketch-based and generation-based methods for Text-to-SQL, improving syntactic accuracy and efficiency. A value-prediction module for WHERE clauses and a new syntax-error evaluation metric further enhance performance.
24	Redefining text to SQL metrics by incorporating semantic and structural similarity	Giovanni Pinna , Yuriy Perezhohin , Luca Manzoni , Mauro Castelli , Andrea De Lorenzo	2025	Link	Text-to-sql	This study introduces a new evaluation metric that more accurately measures SQL query similarity—going beyond simple binary or exact-match scoring by factoring in structural differences, partial correctness, and semantic alignment. The metric allows finer-grained evaluation and better ranking of text-to-SQL models, aiding both model assessment and development.
25	Towards Understanding the Generalization of Medical Text-to-SQL Models and Datasets	Richard Tarbell, Kim-Kwang Raymond Choo, Glenn Dietrich, Anthony Rios	2024	Link	Text-to-sql	The study shows that while current models perform well on existing medical Text-to-SQL benchmarks (with accuracies over 90%), their performance drops sharply (to around 28%) when evaluated on more generalizable splits. The authors introduce a data augmentation technique to improve robustness and highlight that the medical domain still needs more reliable Text-to-SQL solutions
26	A Heterogeneous Graph to Abstract Syntax Tree Framework for Text-to-SQL	Ruisheng Cao, Lu Chen, Jieyu Li, Hanchong Zhang, Hongshen Xu, Wangyou Zhang, Kai Yu	2023	Link	Text-to-sql	The authors propose HG2AST , which encodes input as a heterogeneous graph and decodes into SQL via an abstract syntax tree (AST). On the encoder side, LGESQL performs dual graph message passing; on the decoder side, a grammar-based structure with a learning algorithm controls node expansion order to reduce bias. Together, this unified framework achieves superior performance across diverse text-to-SQL datasets.
27	ExSPIN: Explicit Feedback-Based Self-Play Fine-Tuning for Text-to-	Liang Yan, Jinhang Su, Chuanyi Liu, Shaoming Duan, Yuhao	2025	Link	Text-to-sql	Introduces ExSPIN , a self-play fine-tuning method that incorporates explicit SQL execution feedback and schema hints, significantly improving performance on real-world text-to-

	SQL Parsing	Zhang, Jianhang Li, Peiyi Han, Ye Liu				SQL datasets.
28	An interaction-modeling mechanism for context-dependent Text-to-SQL translation based on heterogeneous graph aggregation	Wei Yu, Tao Chang, Xiaoting Guo, Mengzhu Wang, Xiaodong Wang	2021	Link	Text-to-sql	The authors propose a multi-turn, context-aware Text-to-SQL model where each input (natural language question, previous SQL, and database schema) is represented as a separate graph. They then aggregate these via heterogeneous graph aggregation into a unified representation, resulting in competitive performance on SparC and CoSQL benchmarks.
29	SEOSS-Queries - a software engineering dataset for text-to-SQL and question answering tasks	Mihaela Todorova Tomova, Martin Hofmann, Patrick Mäder	2022	Link	Text-to-sql	Introduces SEOSS-Queries , a software engineering-focused Text-to-SQL dataset containing 1,162 natural language utterances mapped to 166 SQL queries. Each query includes four precise and three more general paraphrased utterances, plus over 393,000 labeled comments extracted from issue trackers. The dataset provides SQLNet and RatSQL baselines for benchmarking.
30	MedT5SQL: a transformers-based large language model for text-to-SQL conversion in the healthcare domain	Alaa Marshan, Anwar Nais Almutairi, Athina Ioannou, David Bell, Asmat Monaghan, Mahir Arzoky	2024	Link	Text-to-sql	Introduces MedT5SQL , a fine-tuned T5 model trained on the MIMICSQL healthcare dataset. It demonstrates effective natural language to SQL conversion in the healthcare domain, showing improved performance with additional training epochs.
31	Yale University's Spider 1.0 NLP Dataset	11 Yale students		Link	Spider Dataset	Spider Dataset

21 → Not about text-to-sql

22 → Not about text-to-sql

24 → About measuring text-to-sql, not developing it.